

Software Architecture - Architectural Styles

Time: 15 minutes.

* Indicates required question

1. Email *

2. Full name *

3. Student ID *

4. An architectural style consists of ____.

1 point

Check all that apply.

- ☐ The vocabulary of components and connectors.
- ☐ A set of constraints on how they can be combined.
- ☐ An informal description of the benefits and drawbacks of using that style.

5. A specific architectural style is chosen for a project because _____. * 1 point

Mark only one oval.

- ☐ The architect has the most familiarity with it.
- ☐ It is known to ensure quality attributes of concern.
- ☐ It will make the system easier to maintain.
- ☐ The developers who will implement the system are most familiar with it.

6. If you have to use a programming language that is largely procedural, which architecture style are you likely to use? * 1 point

Mark only one oval.

- ☐ Data Abstraction and Object-Oriented Organization.
- ☐ Main Program and Subroutines.
- ☐ Event-Based Style.
- ☐ Pipes and Filters Style.

7. What are **two** disadvantages of the repository style? * 1 point

Check all that apply.

- ☐ The existing data schema is difficult to change.
- ☐ The system becomes heavily reliant on the central data.
- ☐ Considerable overhead for data transfer between data accessors.
- ☐ Data integrity can be compromised by having it widely accessible.

8. Which architectural layers can be combined into one layer in small applications?

* 1 point

Mark only one oval.

- ☐ Presentation and Business Logic layers.
- ☐ Business Logic and Data Access layers.
- ☐ Data Access and Database layers.

9. Which architectural layers can be deployed into one tier in small monolith applications?

* 1 point

Mark only one oval.

- ☐ Presentation, Business Logic and Data Access layers.
- ☐ Business Logic and Data Access layers.
- ☐ Presentation and Business Logic layers.

10. Which of these is an advantage of the event-based style? *

1 point

Mark only one oval.

- ☐ Events trigger responses in a predictable manner.
- ☐ All interactions happen synchronously.
- ☐ Events are processed with more efficiency.
- ☐ Event generators and event consumers are loosely coupled.

11. Which of these applications would be best suited to the pipes and filters style? * 1 point

Mark only one oval.

- ☐ A self-driving car.
- ☐ Analyzing data and producing reports.
- ☐ Accessing data from a repository.
- ☐ A user interface, such as a simple poker game.

12. What are the **three** advantages of the pipes and filters style? * 1 point

Check all that apply.

- ☐ Filters are loosely coupled.
- ☐ Filters can be reused.
- ☐ Filters provide a complete transformation of input data to output data.
- ☐ Filter can be implemented as a separate task and potentially executed in parallel with other filters.

13. What are the **three** advantages of the interpreter style? * 1 point

Check all that apply.

- ☐ Allowing developers to develop add-ons in a common language.
- ☐ Abstracting away platform details.
- ☐ Giving end users the opportunity to program scripts or macros.
- ☐ Enhancing resource usage efficiency.

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